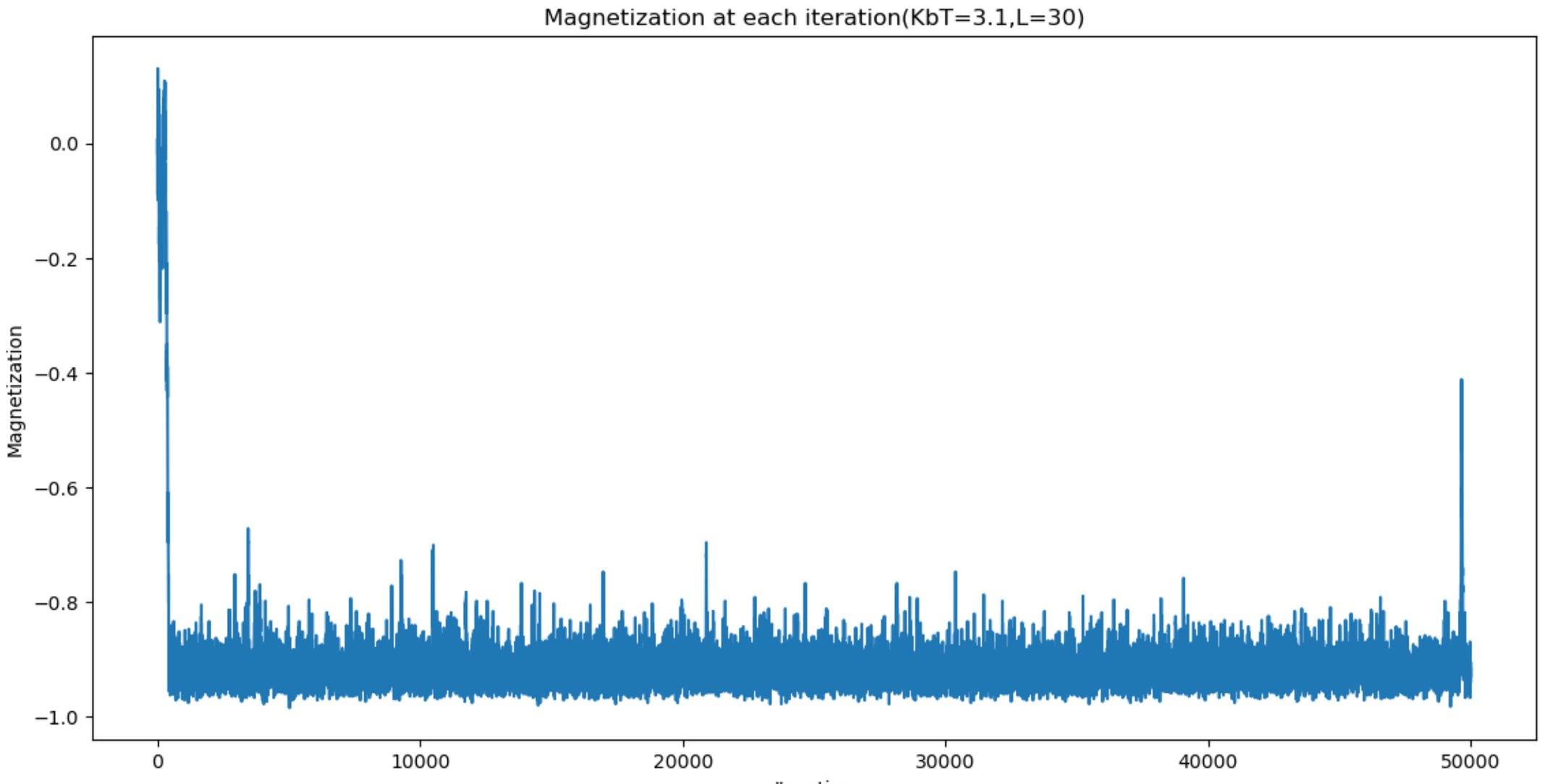
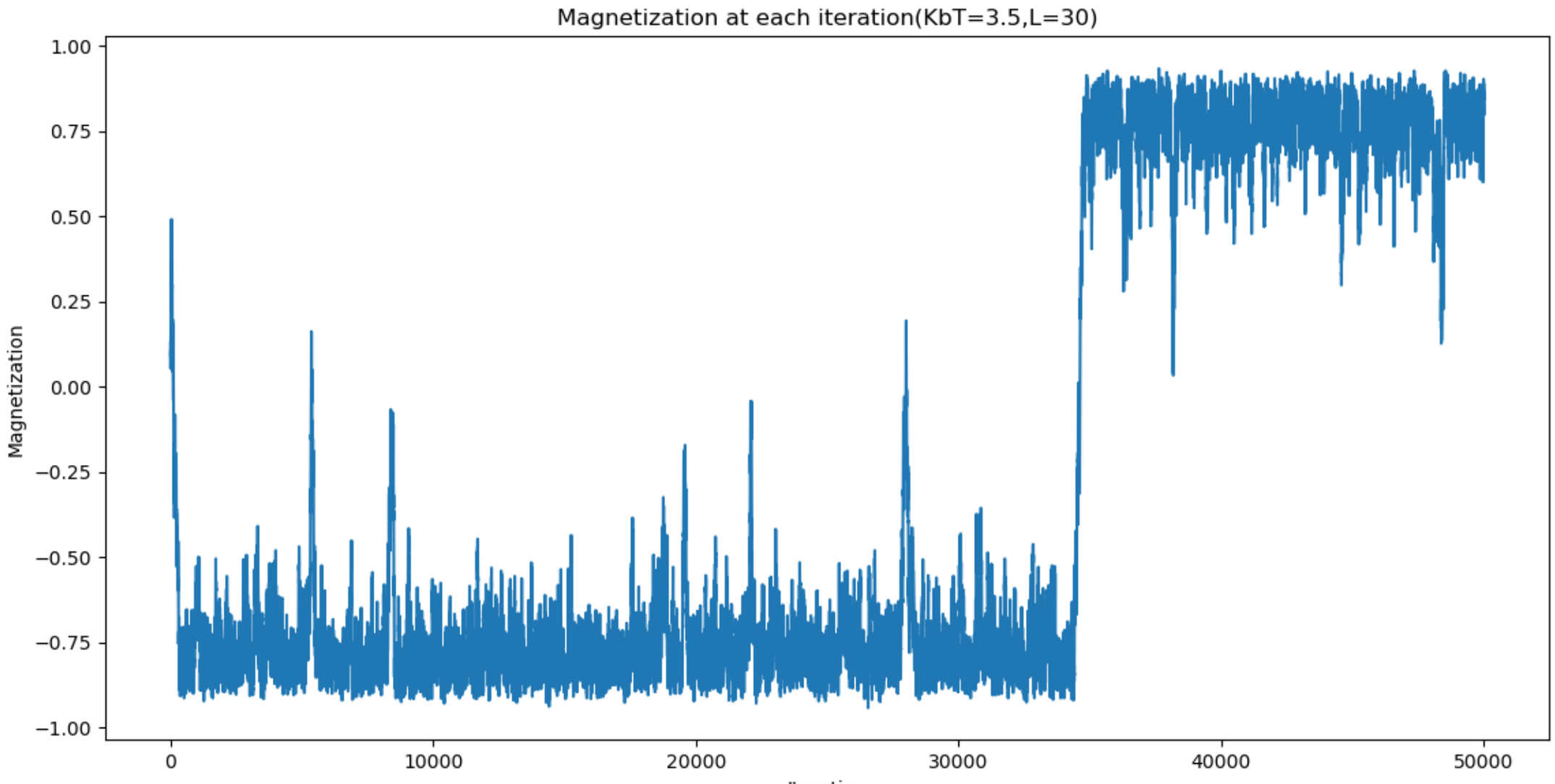
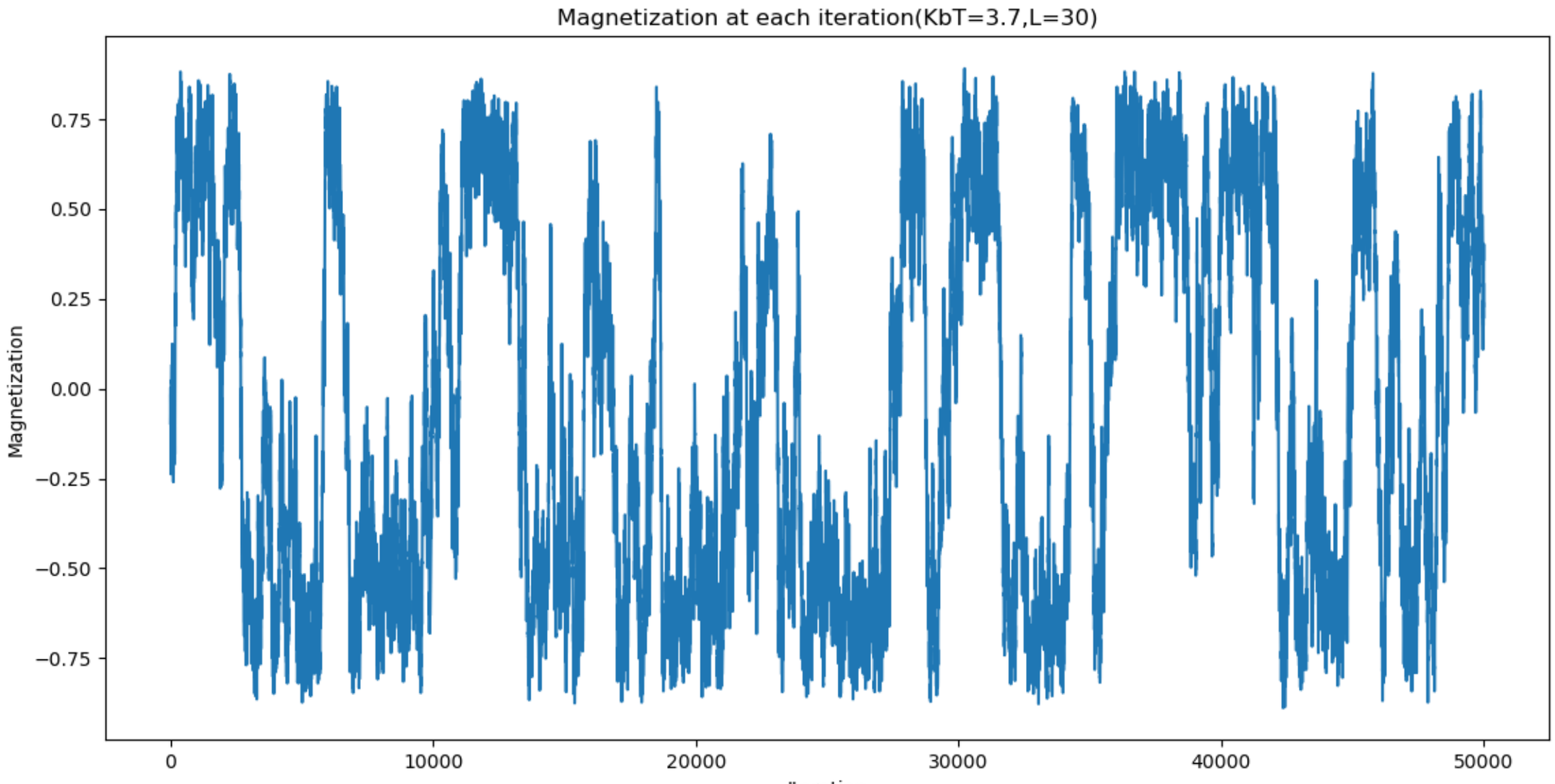
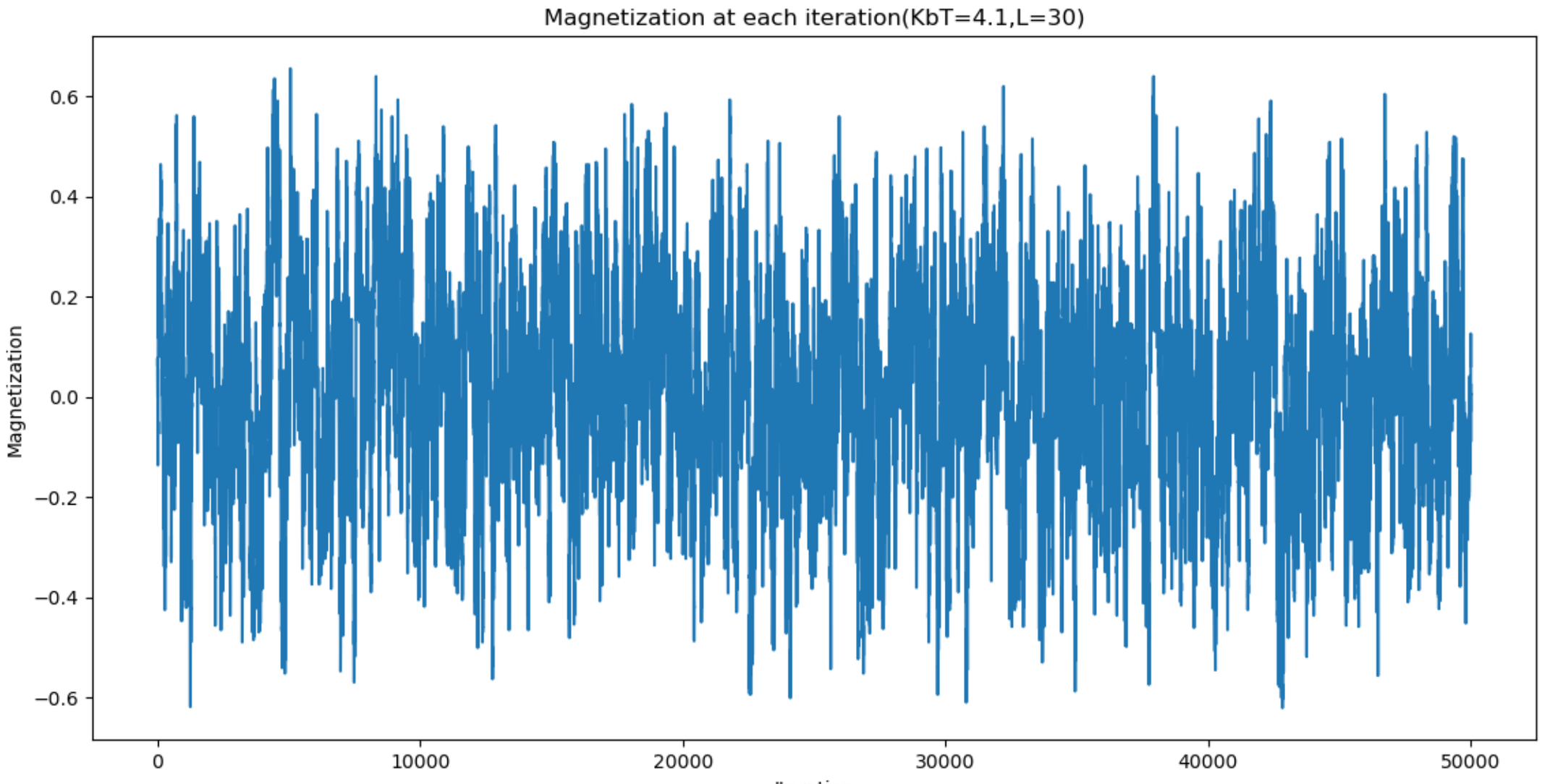
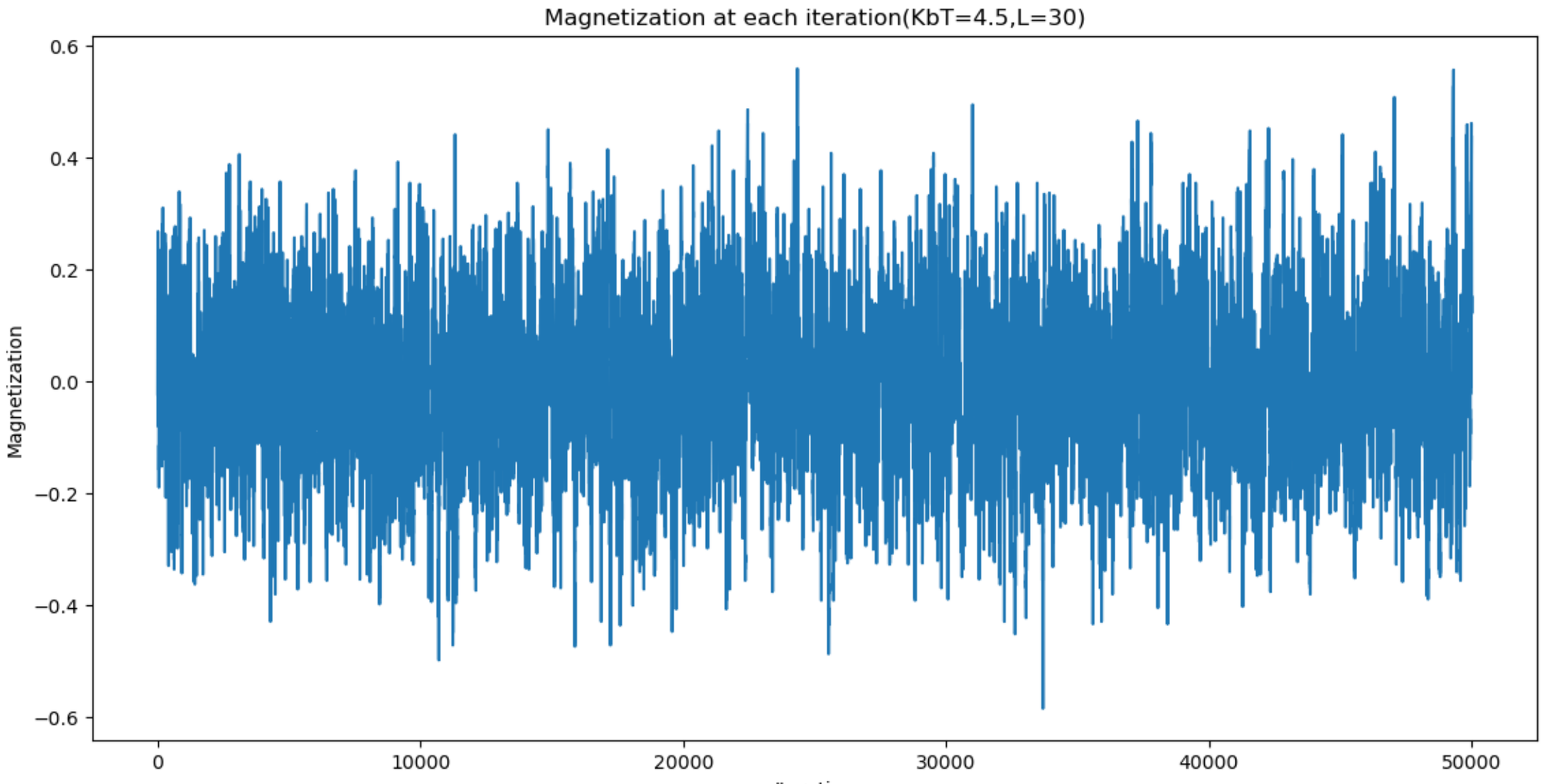
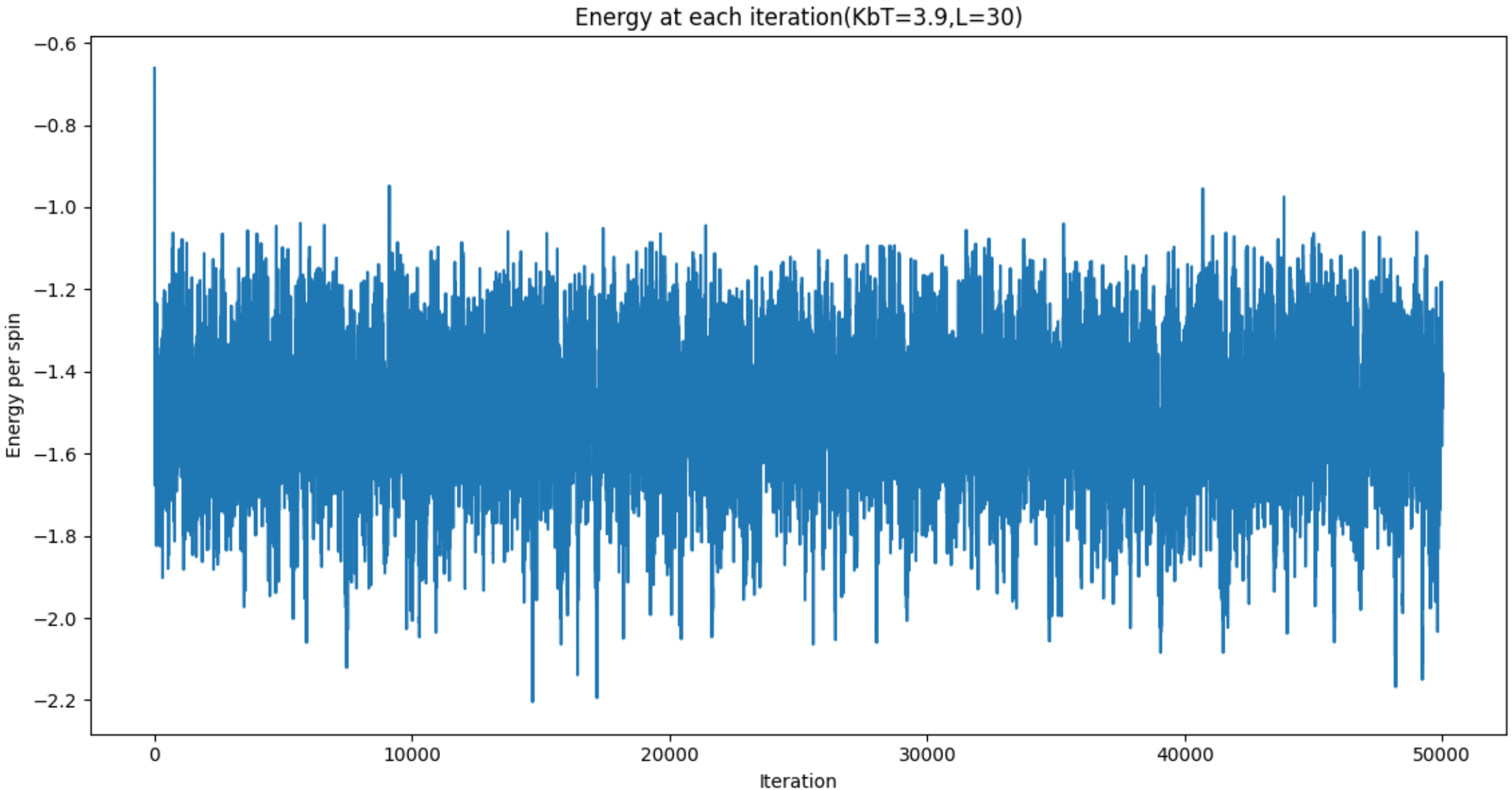

Magnetization at each iteration(KbT=3.1,L=30)
Magnetization
0.0
−0.2
−0.4
−0.6
−0.8
−1.0
0
10000
20000
30000
40000
50000
Iteration

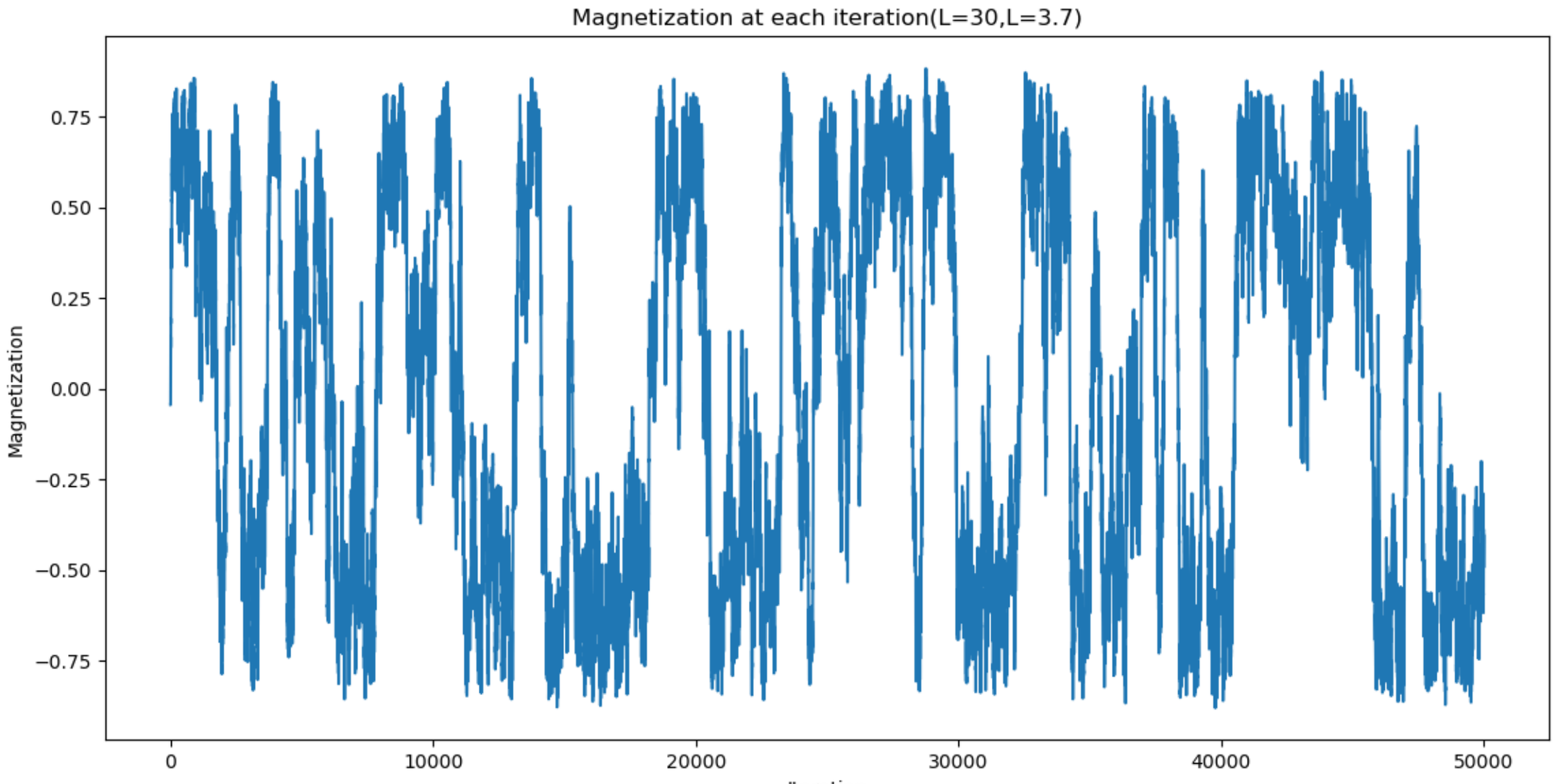

Magnetization at each iteration(KbT=3.5,L=30)
Magnetization
Iteration
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10000
20000
30000
40000
50000

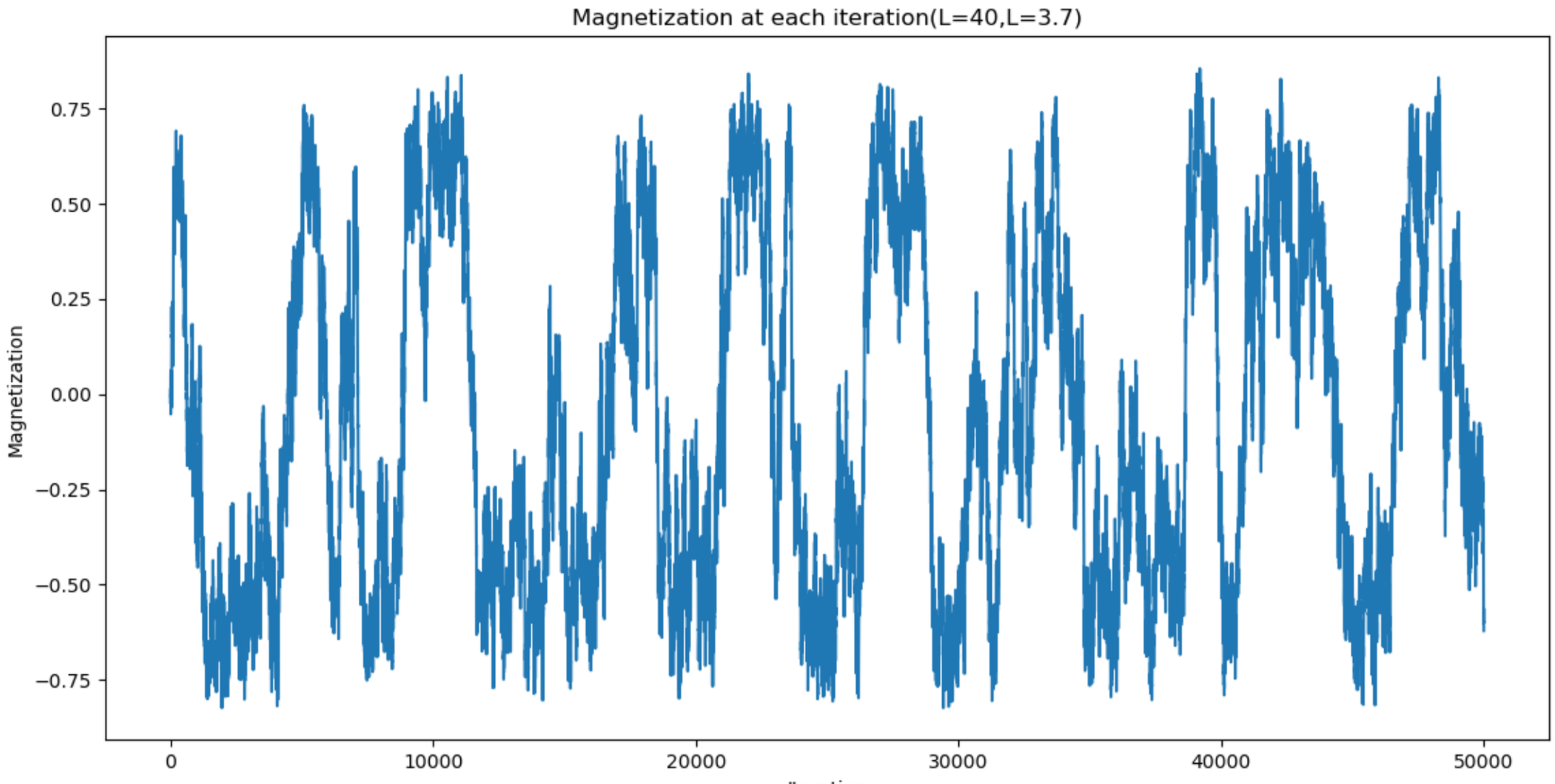

Magnetization at each iteration(KbT=3.7,L=30)
Magnetization
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0
10000
20000
30000
40000
50000
Iteration

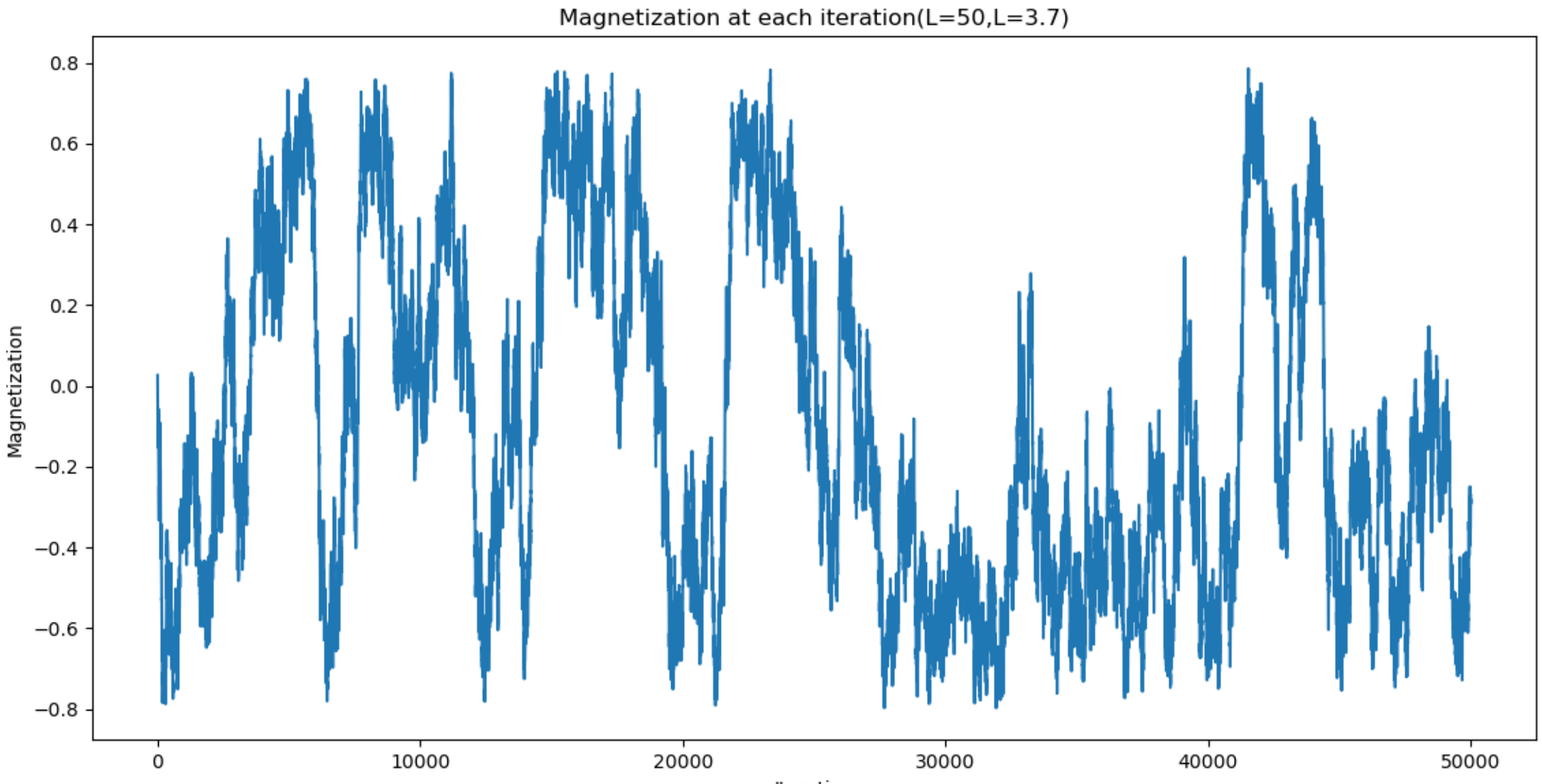

Magnetization at each iteration(KbT=4.1,L=30)
Magnetization
Iteration
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
10000
20000
30000
40000
50000

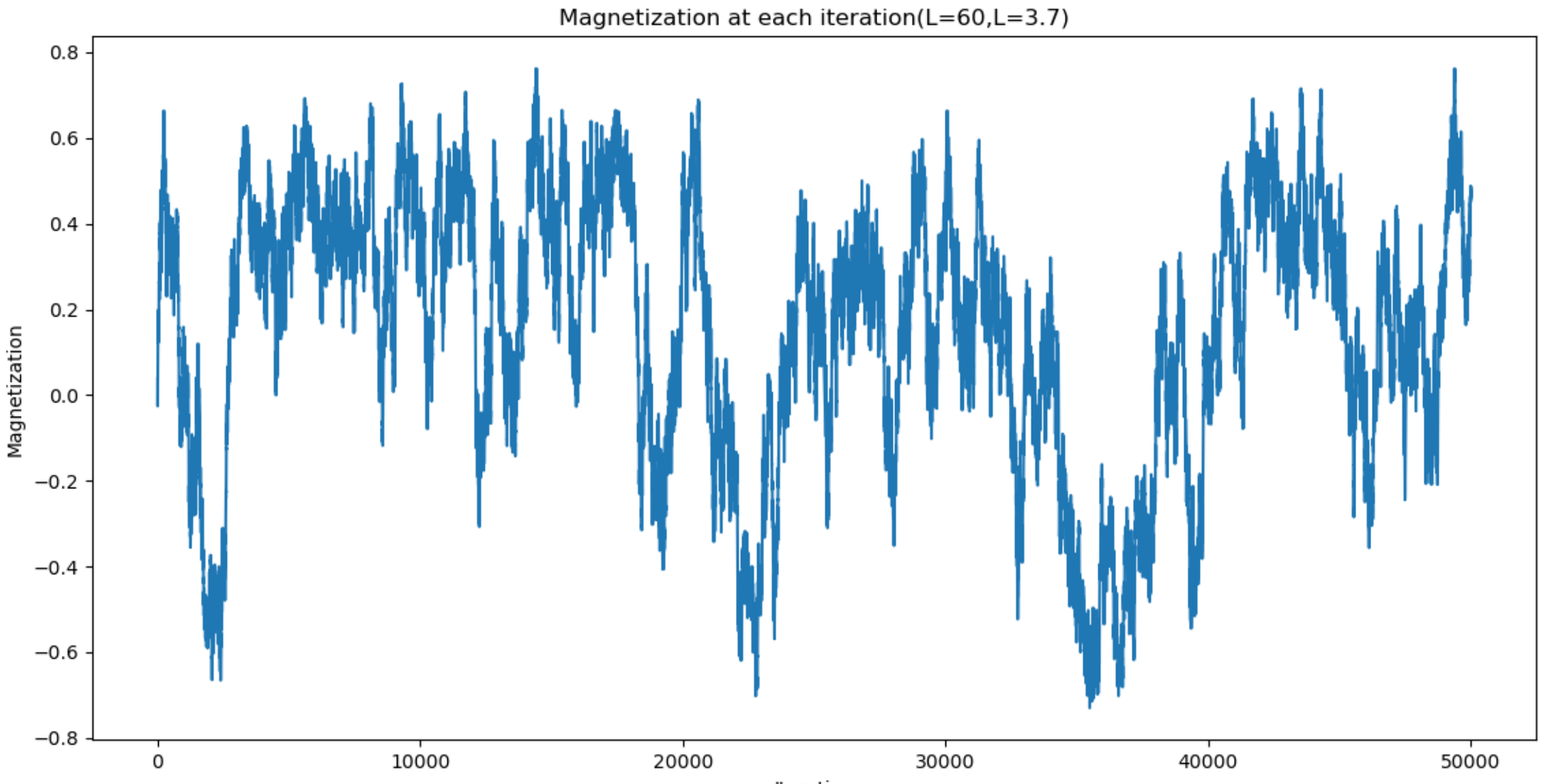

Magnetization at each iteration(KbT=4.5,L=30)
Magnetization
Iteration
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
10000
20000
30000
40000
50000

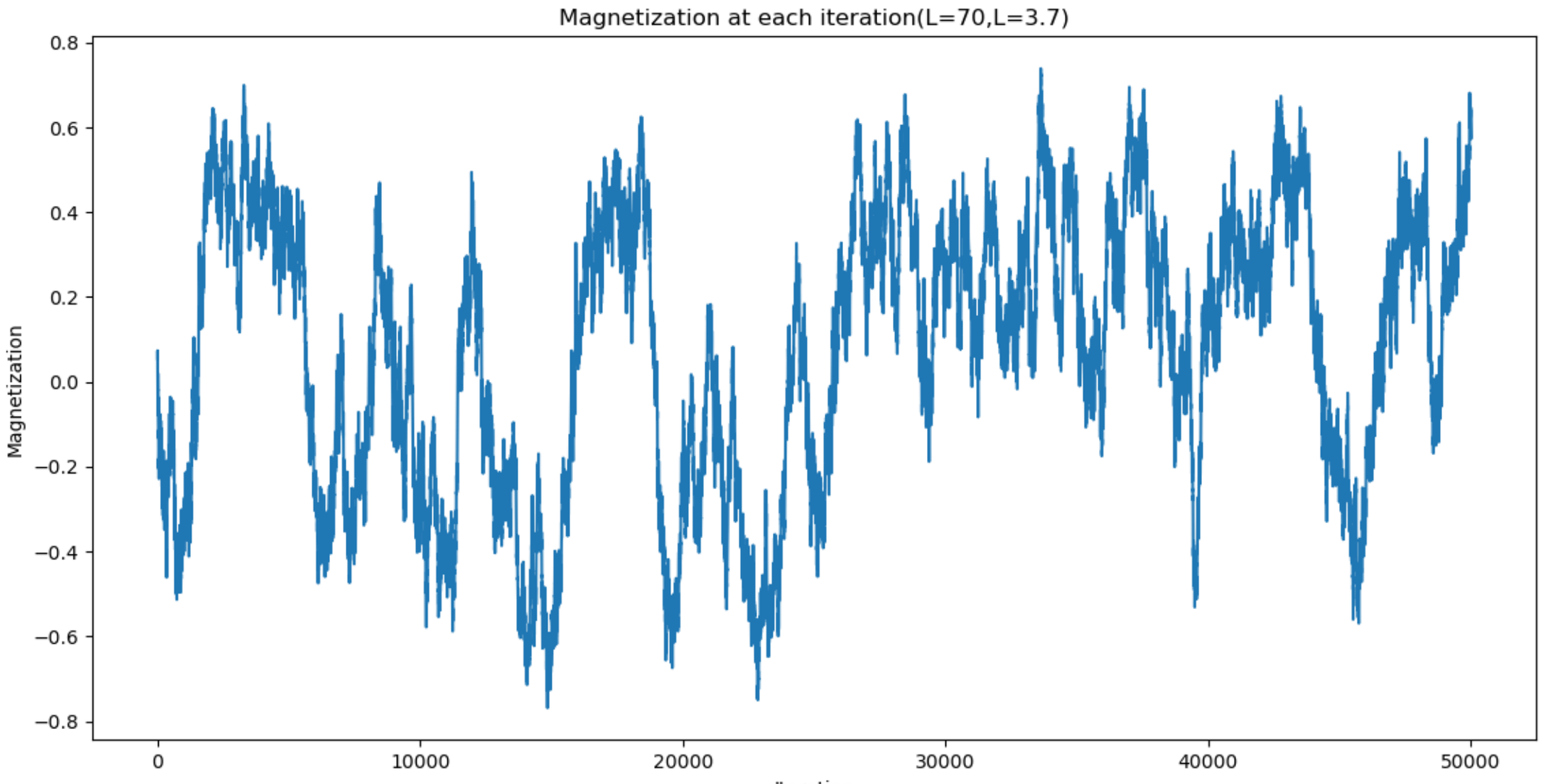

Energy at each iteration(KbT=3.9,L=30)
Energy per spin
Iteration
−0.6
−0.8
−1.0
−1.2
−1.4
−1.6
−1.8
−2.0
−2.2
0
10000
20000
30000
40000
50000

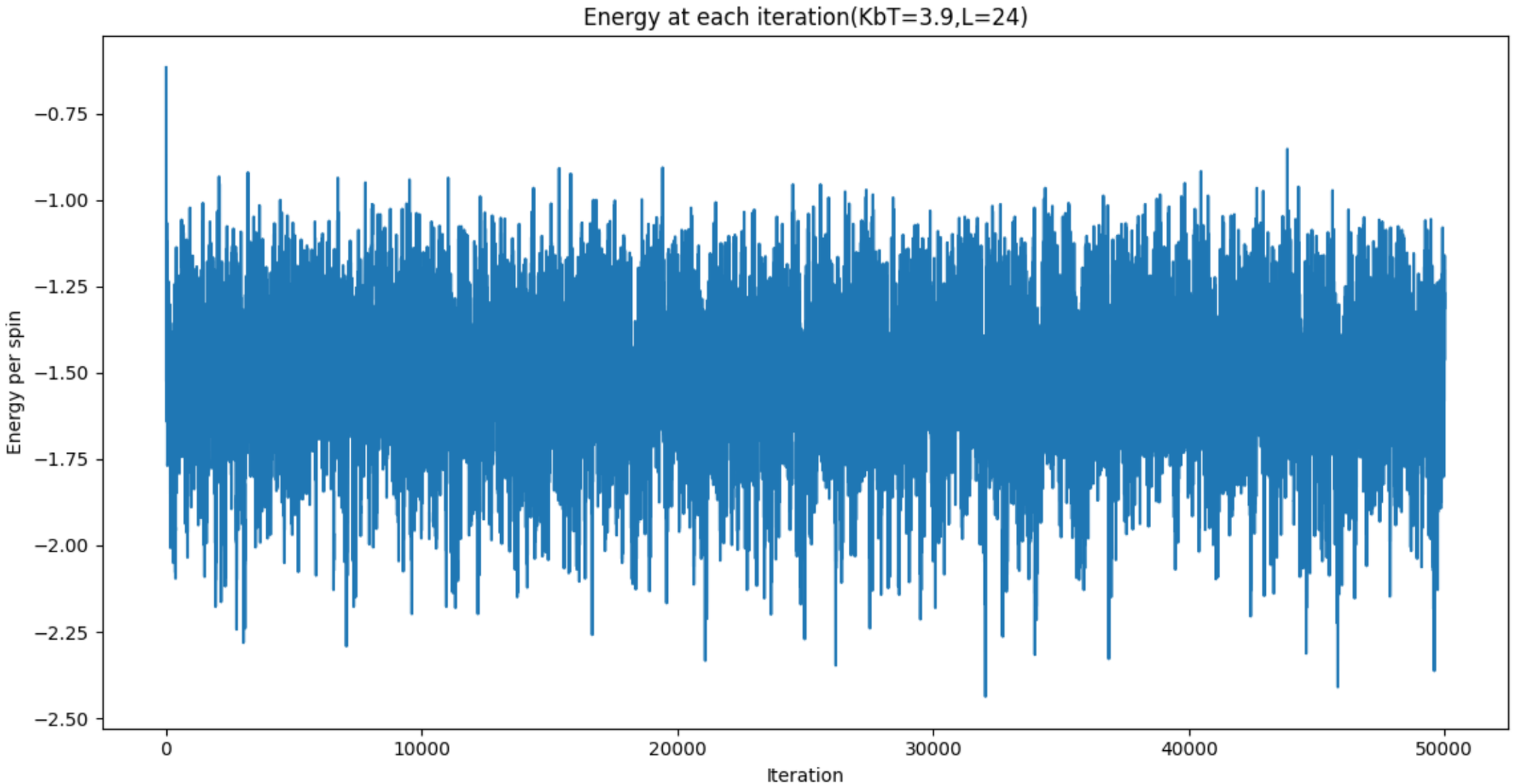

Magnetization at each iteration(L=30,L=3.7)
Magnetization
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0
10000
20000
30000
40000
50000
Iteration

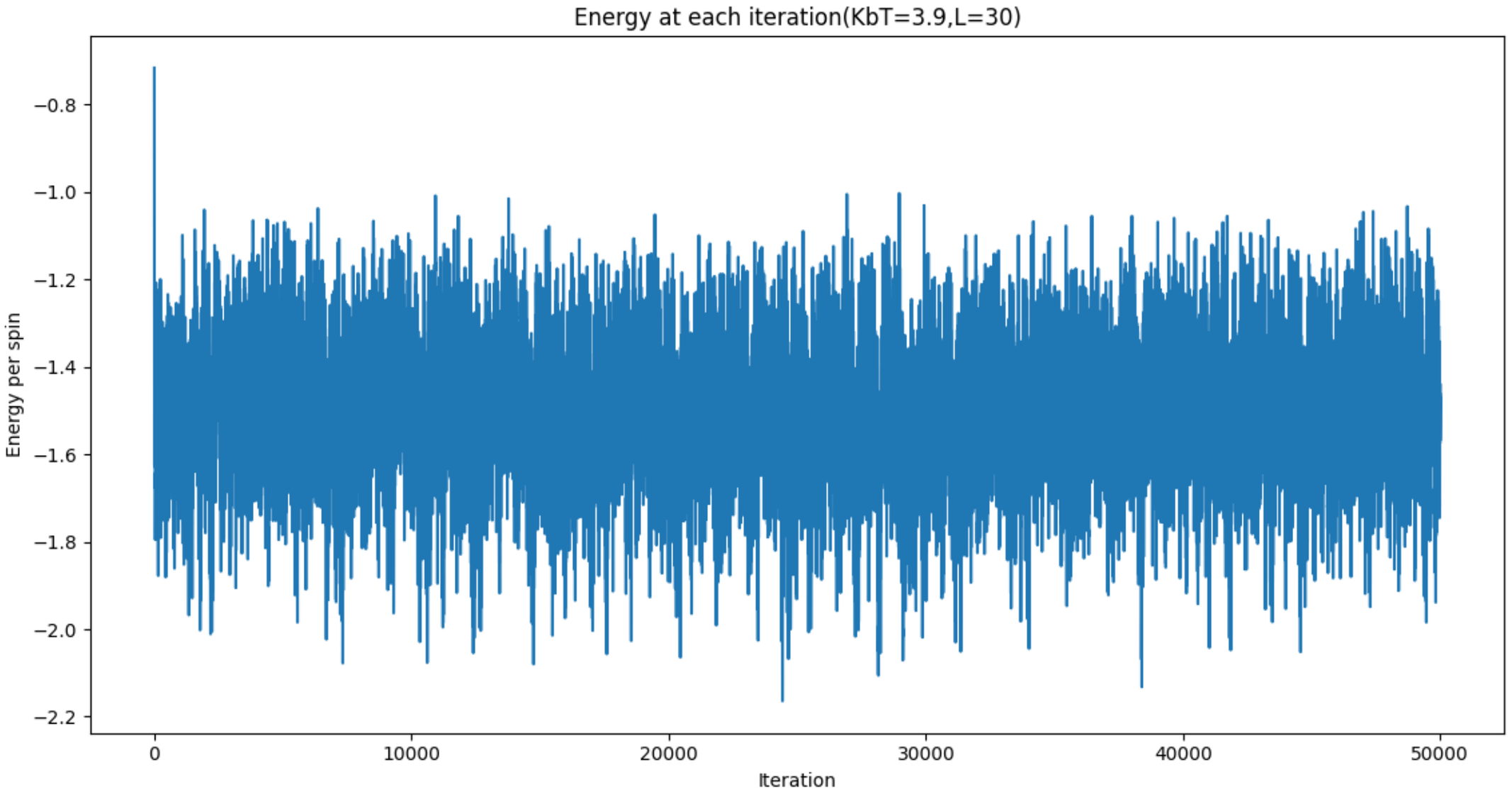

Magnetization at each iteration(L=40,L=3.7)
Magnetization
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
0
10000
20000
30000
40000
50000
Iteration

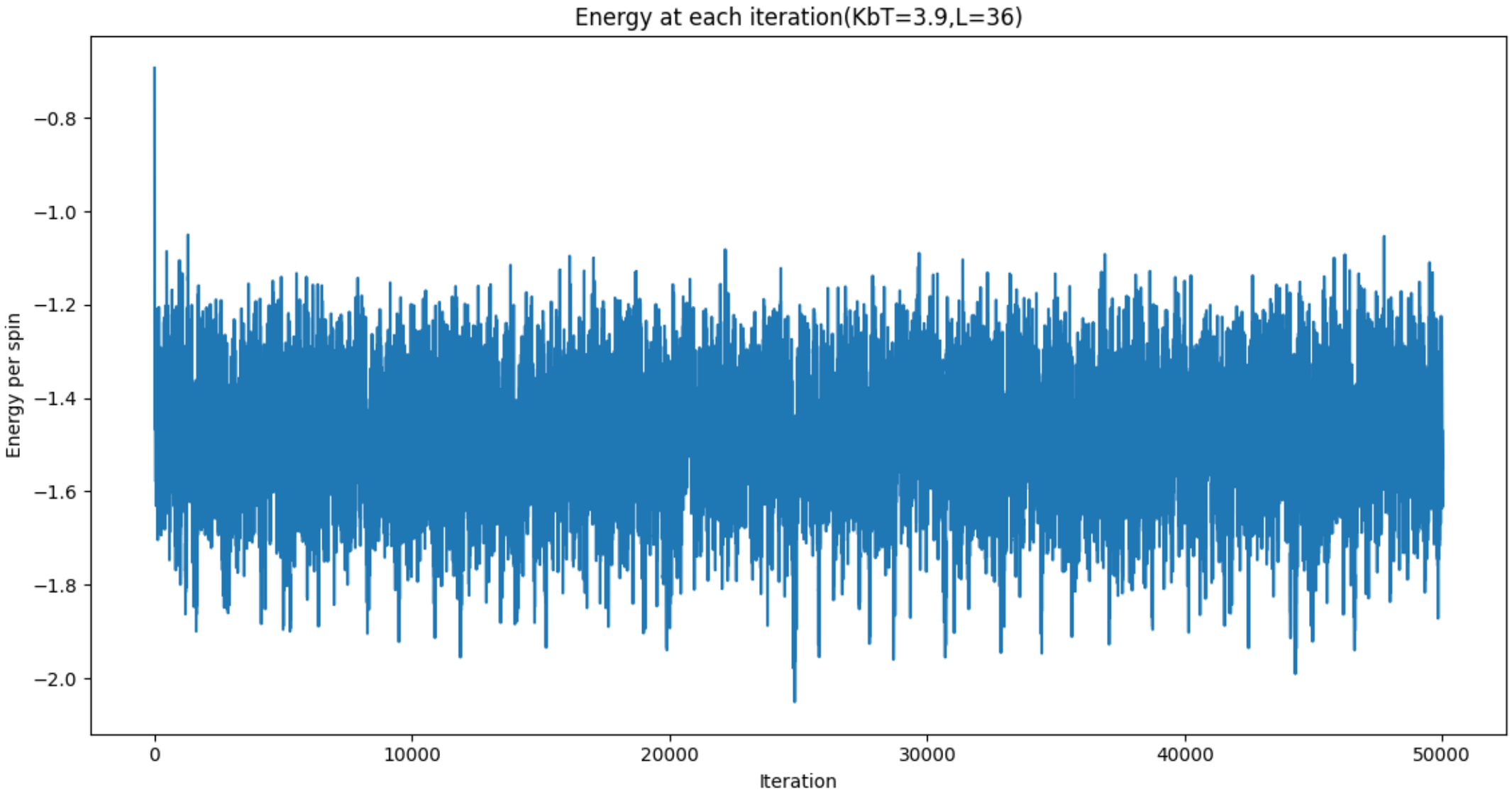

Magnetization at each iteration(L=50,L=3.7)
Magnetization
Iteration
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0
10000
20000
30000
40000
50000

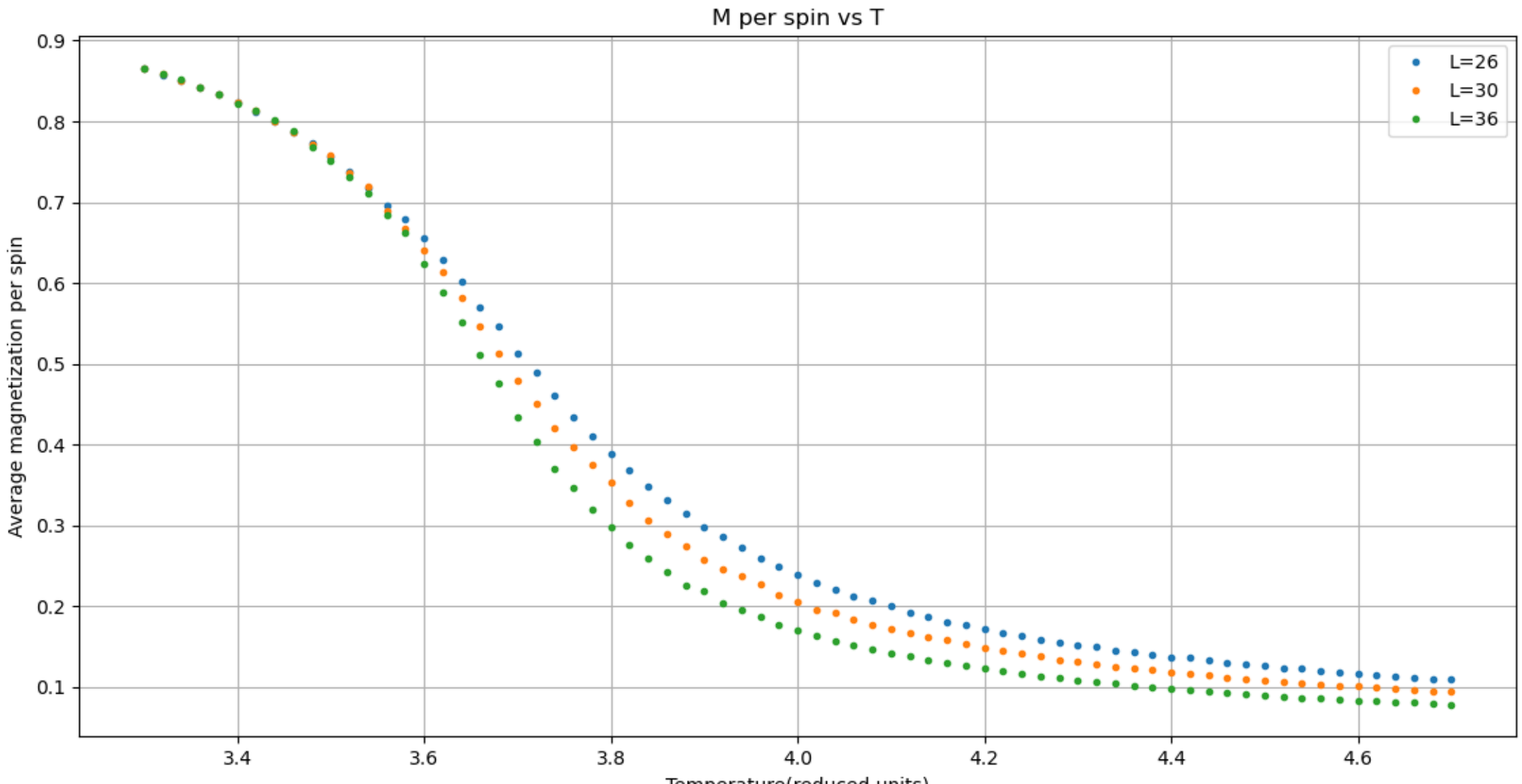

Magnetization at each iteration(L=60,L=3.7)
Magnetization
Iteration
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0
10000
20000
30000
40000
50000

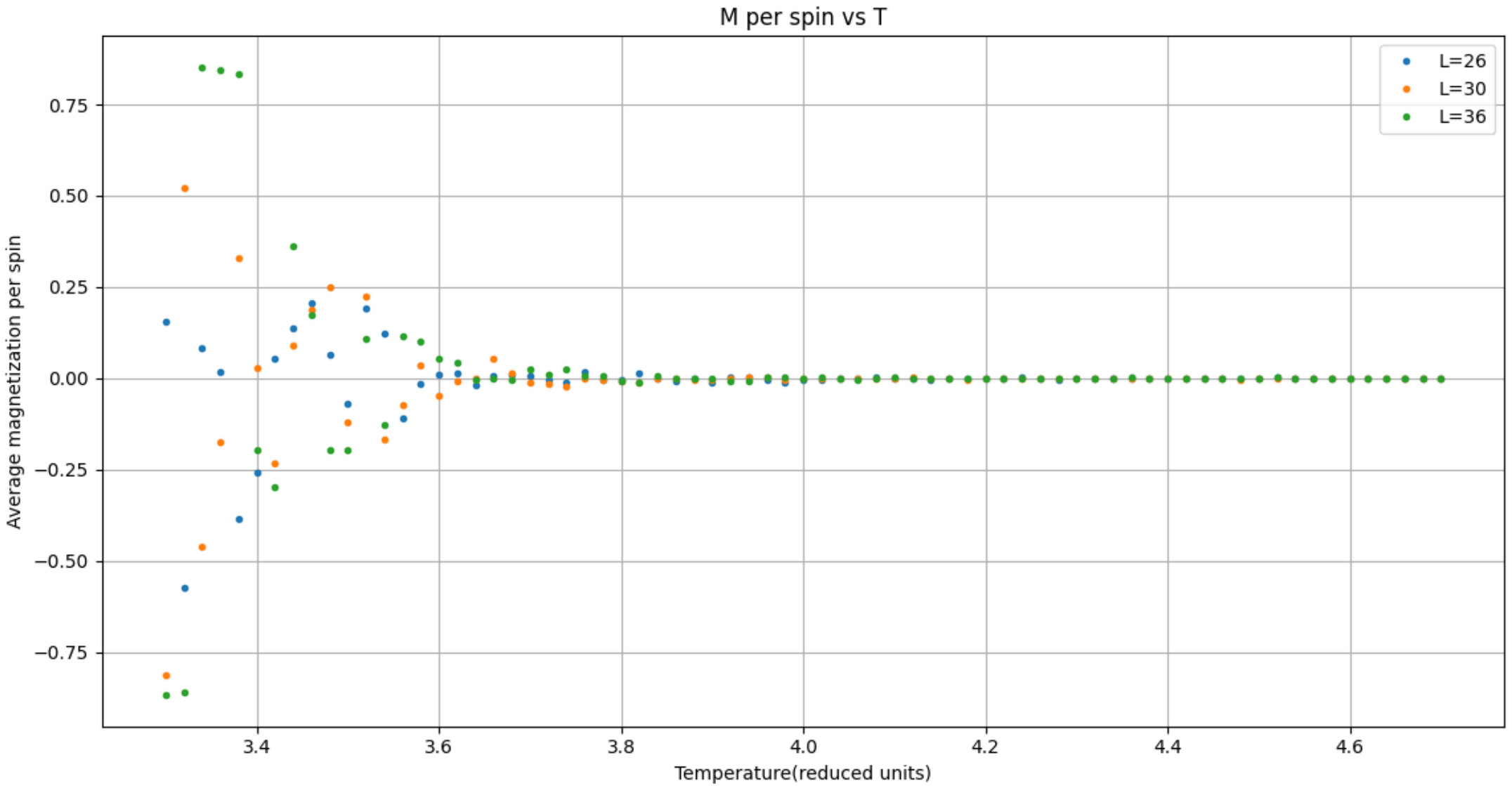

Magnetization at each iteration(L=70,L=3.7)
Magnetization
Iteration
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
−0.8
0
10000
20000
30000
40000
50000

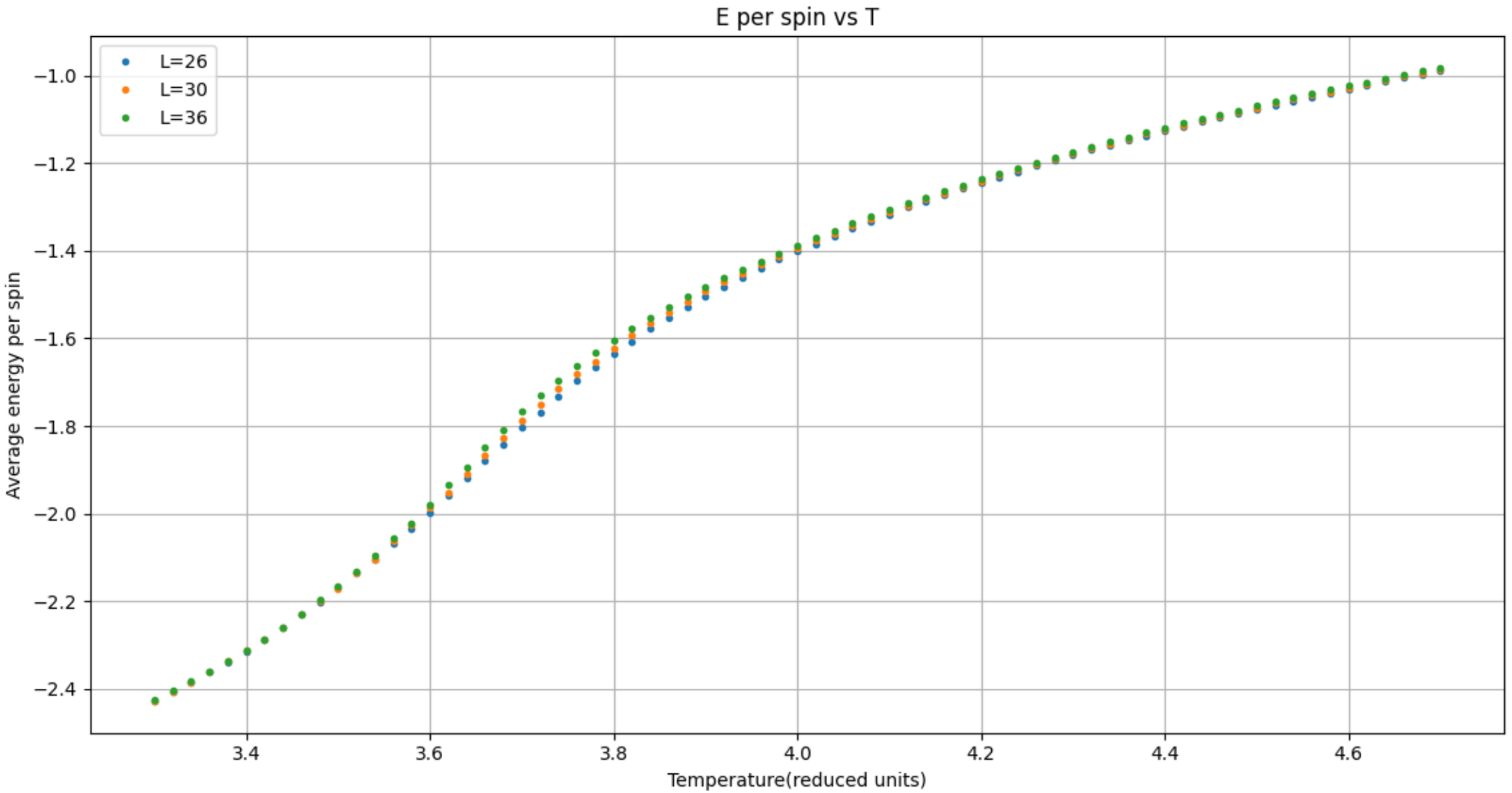

Energy at each iteration(KbT=3.9,L=24)
Energy per spin
−0.75
−1.00
−1.25
−1.50
−1.75
−2.00
−2.25
−2.50
0
10000
20000
30000
40000
50000
Iteration

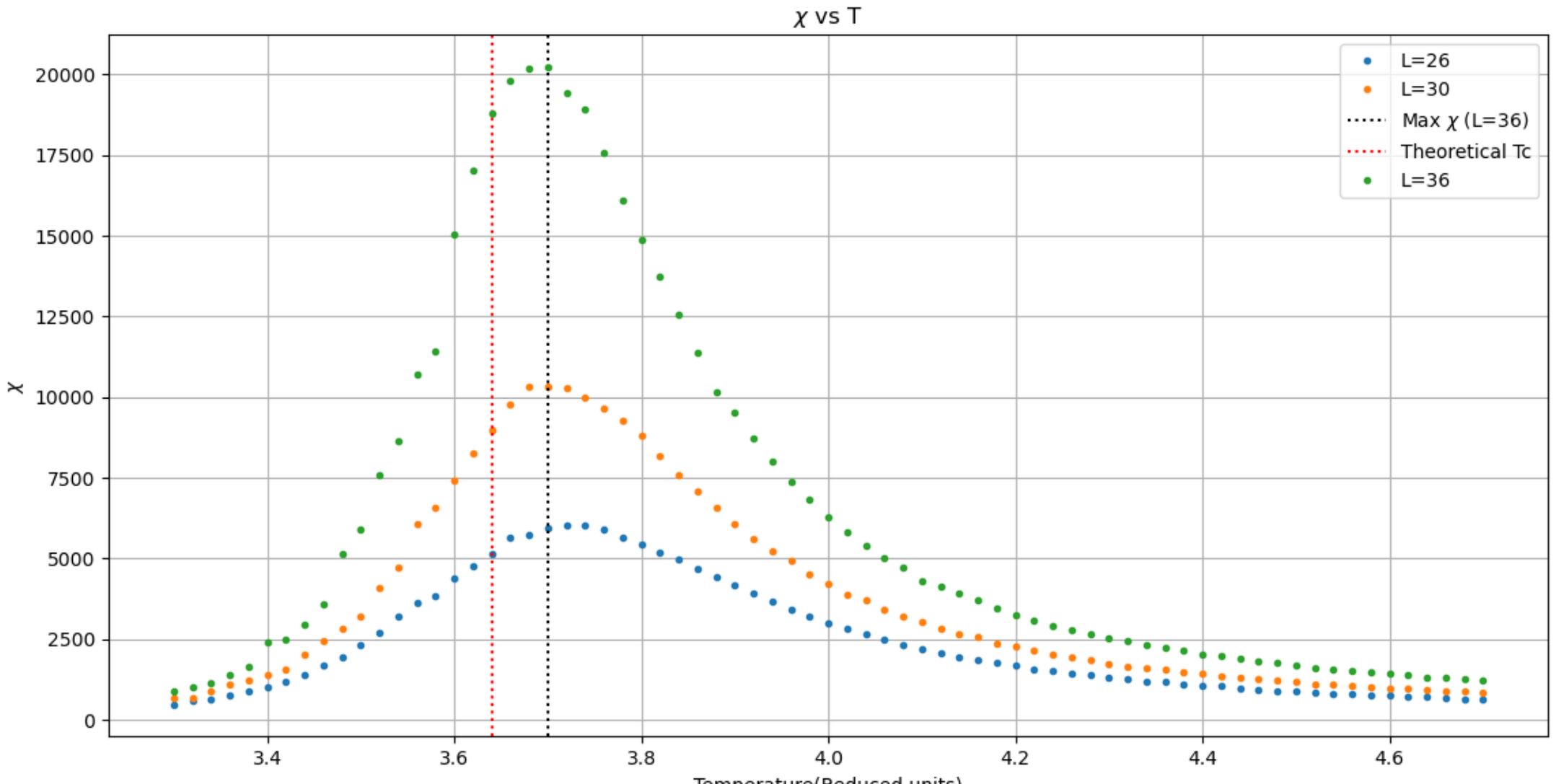

Energy at each iteration(KbT=3.9,L=30)
Energy per spin
−0.8
−1.0
−1.2
−1.4
−1.6
−1.8
−2.0
−2.2
0
10000
20000
30000
40000
50000
Iteration

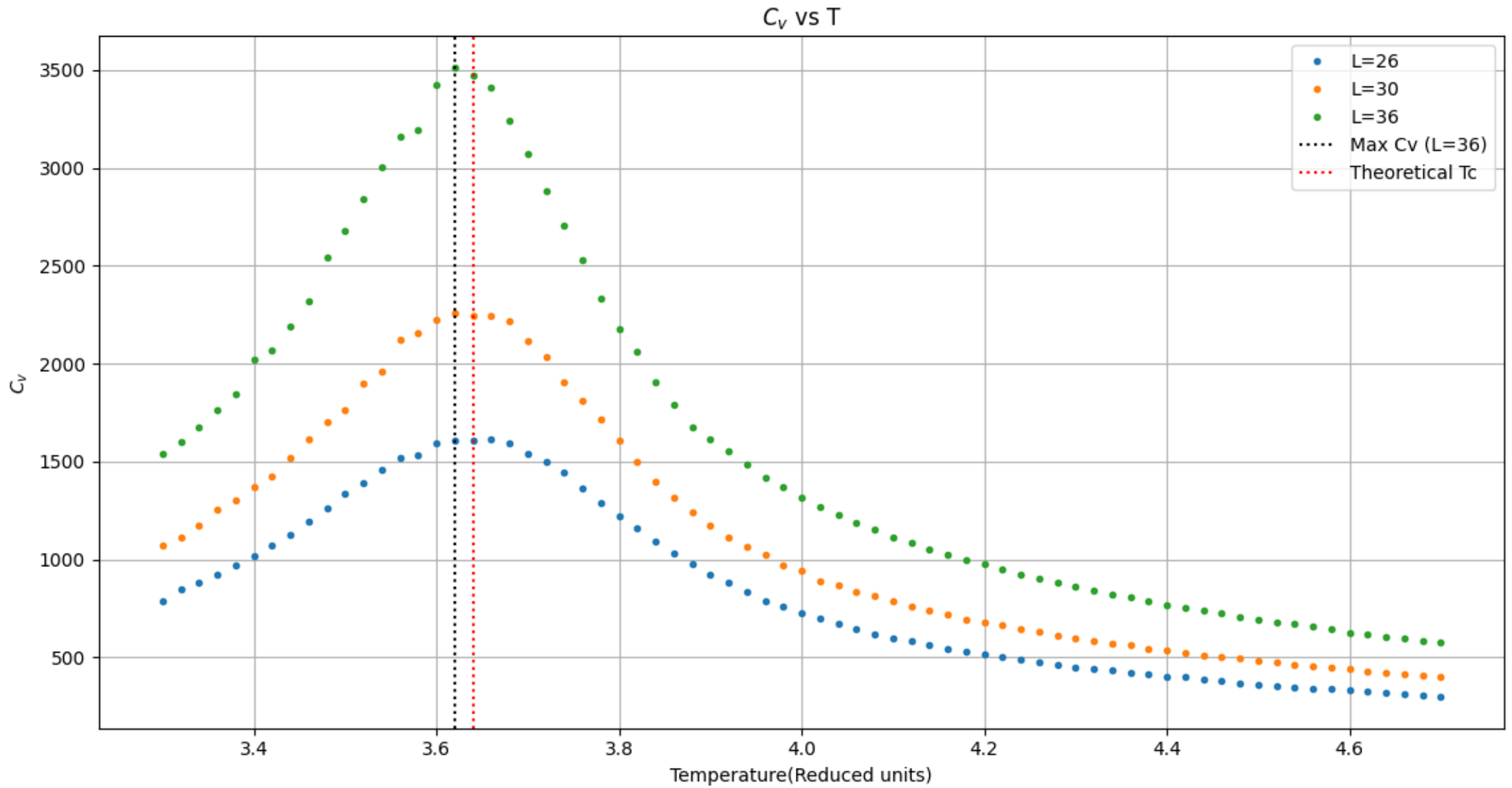

Energy at each iteration(KbT=3.9,L=36)
Energy per spin
−0.8
−1.0
−1.2
−1.4
−1.6
−1.8
−2.0
0
10000
20000
30000
40000
50000
Iteration

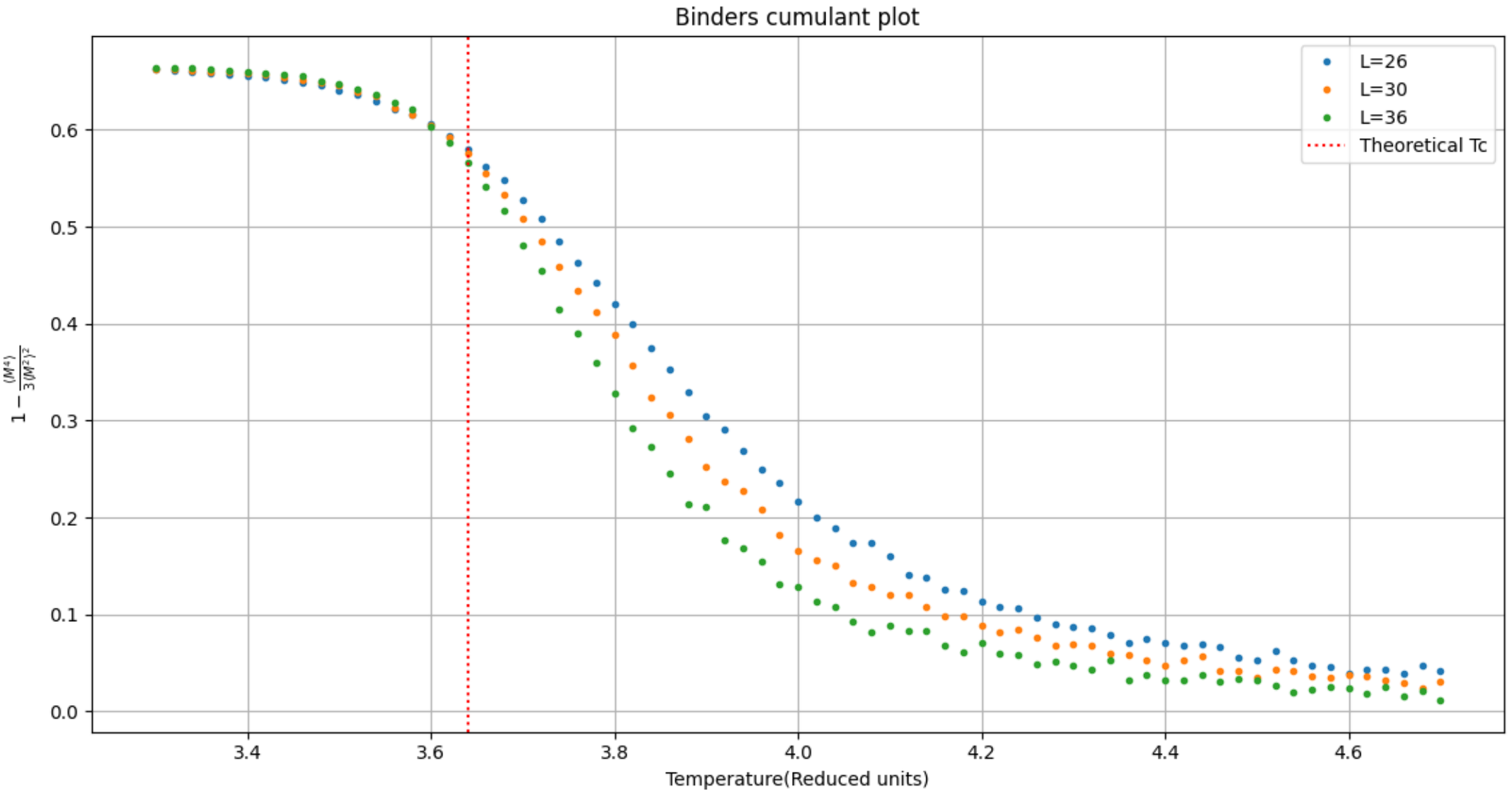

M per spin vs T
L=26
L=30
L=36
Average magnetization per spin
Temperature(reduced units)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
0.1
3.4
3.6
3.8
4.0
4.2
4.4
4.6


M per spin vs T
L=26
L=30
L=36
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Average magnetization per spin
3.4
3.6
3.8
4.0
4.2
4.4
4.6
Temperature(reduced units)


E per spin vs T
L=26
L=30
L=36
−1.0
−1.2
−1.4
−1.6
−1.8
−2.0
−2.2
−2.4
Average energy per spin
3.4
3.6
3.8
4.0
4.2
4.4
4.6
Temperature(reduced units)


χ vs T
L=26
L=30
Max χ (L=36)
Theoretical Tc
L=36
χ
Temperature(Reduced units)
0
2500
5000
7500
10000
12500
15000
17500
20000
3.4
3.6
3.8
4.0
4.2
4.4
4.6


C_v vs T
L=26
L=30
L=36
Max Cv (L=36)
Theoretical Tc
3500
3000
2500
2000
1500
1000
500
C_v
3.4
3.6
3.8
4.0
4.2
4.4
4.6
Temperature(Reduced units)


Binders cumulant plot
L=26
L=30
L=36
Theoretical Tc
0.0
0.1
0.2
0.3
0.4
0.5
0.6
3.4
3.6
3.8
4.0
4.2
4.4
4.6
Temperature(Reduced units)
$1 - \frac{\langle M^4 \rangle}{3\langle M^2 \rangle^2}$